



Welcome to DP Physics SL/HL

BROUGHT TO YOU BY MR. KNIGHT

What is Physics?



- ▶ Physics is the study of...
 - ▶ Energy
 - ▶ Matter
 - ▶ ... and their interaction

Why did you sign up for physics?

What is DP Physics?

- ▶ Over the next ~2 years, we will study 5 Topics that cover a wide range of modern physics.
 - ▶ You can expect 2 tests to take place per unit.
 - ▶ One mid-way through the unit, and another at the end.
 - ▶ You will also complete 2 cumulative mock exams and 1 final exam during the dedicated exam sessions.
- ▶ The Internal Assessment (“IA”) is an independent investigation designed, carried out, and evaluated by each student.
- ▶ Class time will be given for review before all assessments.
 - ▶ Typically, 1 review class before unit tests
 - ▶ At least 1 week of classes before mock/final exams.

Syllabus component	Teaching hours	
	SL	HL
Syllabus content	110	180
A. Space, time and motion	27	42
B. The particulate nature of matter	24	32
C. Wave behaviour	17	29
D. Fields	19	38
E. Nuclear and quantum physics	23	39
Experimental programme	40	60
Practical work	20	40
Collaborative sciences project	10	10
Scientific investigation	10	10
Total teaching hours	150	240

Distinction between SL and HL

Students at SL and HL share the following.

- An understanding of science through a stimulating experimental programme
- The nature of science as an overarching theme
- The study of a concept-based syllabus
- One piece of internally assessed work, the scientific investigation
- The collaborative sciences project

The SL course provides students with a fundamental understanding of physics and experience of the associated skills. The HL course requires students to increase their knowledge and understanding of the subject, and so provides a solid foundation for further study at university level.

The SL course has a recommended 150 teaching hours, compared to 240 hours for the HL course. This difference is reflected in the additional content studied by HL students. Some of the HL content is conceptually more demanding and explored in greater depth. The distinction between SL and HL is therefore one of both breadth and depth. The increased breadth and depth at HL result in increased networked knowledge, requiring the student to make more connections between diverse areas of the syllabus.

A. Space, time and motion	B. The particulate nature of matter	C. Wave behaviour	D. Fields	E. Nuclear and quantum physics
A.1 Kinematics • A.2 Forces and momentum • A.3 Work, energy and power • A.4 Rigid body mechanics ... A.5 Galilean and special relativity ...	B.1 Thermal energy transfers • B.2 Greenhouse effect • B.3 Gas laws • B.4 Thermodynamics ... B.5 Current and circuits •	C.1 Simple harmonic motion .. C.2 Wave model • C.3 Wave phenomena .. C.4 Standing waves and resonance • C.5 Doppler effect ..	D.1 Gravitational fields .. D.2 Electric and magnetic fields .. D.3 Motion in electromagnetic fields • D.4 Induction ...	E.1 Structure of the atom .. E.2 Quantum physics ... E.3 Radioactive decay .. E.4 Fission • E.5 Fusion and stars •

- Topics with content that should be taught to all students
- Topics with content that should be taught to all students plus additional HL content
- Topics with content that should only be taught to HL students

Example topic sequence

- A1 → A2 → A3 → D1
- B5 → D2 → D3 → D4
- B1 → B2 → B3 → B4
- C1 → C2 → C3 → C4 → C5
- E1 → E2 → E3 → E4 → E5
- A4 → A5

Table 4

Physics syllabus content overview

A. Space, time and motion	B. The particulate nature of matter	C. Wave behaviour	D. Fields	E. Nuclear and quantum physics
A.1 Kinematics • A.2 Forces and momentum • A.3 Work, energy and power • A.4 Rigid body mechanics ... A.5 Galilean and special relativity ...	B.1 Thermal energy transfers • B.2 Greenhouse effect • B.3 Gas laws • B.4 Thermodynamics ... B.5 Current and circuits •	C.1 Simple harmonic motion .. C.2 Wave model • C.3 Wave phenomena .. C.4 Standing waves and resonance • C.5 Doppler effect ..	D.1 Gravitational fields .. D.2 Electric and magnetic fields .. D.3 Motion in electromagnetic fields • D.4 Induction ...	E.1 Structure of the atom .. E.2 Quantum physics ... E.3 Radioactive decay .. E.4 Fission • E.5 Fusion and stars •

• Topics with content that should be taught to all students

.. Topics with content that should be taught to all students plus additional HL content

... Topics with content that should only be taught to HL students



Data booklet

The IB publishes a *Physics data booklet* which contains electrical symbols, mathematical equations, constants, and physics equations relevant to the course. Students must have access to a copy for the duration of the course so that they can become familiar with its contents. Direct reference is made to relevant equations in the understandings sections of the guide. This helps to maintain the emphasis on interpretation and application rather than memorization of symbols, constants and equations. A clean copy of the *Physics data booklet* must also be made available to candidates for all examination papers at both SL and HL.

Assessment outline—SL

Assessment component	Weighting
External assessment (3 hours)	80%
Paper 1 (1 hour and 30 minutes) Paper 1A—Multiple-choice questions Paper 1B—Data-based questions (Total 45 marks)	36%
Paper 2 (1 hour and 30 minutes) Short-answer and extended-response questions on standard level material only. (Total 50 marks)	44%
Internal assessment (10 hours)	20%
The internal assessment consists of one task: the scientific investigation. This component is internally assessed by the teacher and externally moderated by the IB at the end of the course. (Total 24 marks)	

Assessment outline—HL

Assessment component	Weighting
External assessment (4 hours 30 minutes)	80%
Paper 1 (2 hours) Paper 1A—Multiple-choice questions Paper 1B—Data-based questions (Total 60 marks)	36%
Paper 2 (2 hour and 30 minutes) Short-answer and extended-response questions on standard level and additional higher level material. (Total 90 marks)	44%
Internal assessment (10 hours)	20%
The internal assessment consists of one task: the scientific investigation. This component is internally assessed by the teacher and externally moderated by the IB at the end of the course. (Total 24 marks)	

Paper 1

Duration: 1 hour and 30 minutes

Weighting: 36%

Marks: 45

Paper 1 is presented as two separate booklets

Paper 1A—25 marks

- 25 multiple-choice questions on standard level material only.
No marks are deducted for incorrect answers.

Paper 1B—20 marks

- Data-based questions.

Paper 1A and paper 1B are to be completed together without interruptions.

Paper 1

Duration: 2 hours

Weighting: 36%

Marks: 60

Paper 1 is presented as two separate booklets

Paper 1A—40 marks

- 40 multiple-choice questions on standard level and additional higher level material.
No marks are deducted for incorrect answers.

Paper 1B—20 marks

- Data-based questions.

Paper 1A and paper 1B are to be completed together without interruptions.

Paper 2

Duration: 1 hour and 30 minutes

Weighting: 44%

Marks: 50

- Short-answer and extended-response questions on standard level material only.

The questions on paper 2 test assessment objectives 1, 2 and 3.

The use of calculators is permitted. See the *Calculators guidance for examinations booklet* on the Programme Resource Centre.

Each student must have access to a clean copy of the *Physics data booklet* during the examination. It is the responsibility of the school to download a copy from IBIS or the Programme Resource Centre and to ensure that there are sufficient copies available for all students.

Paper 2

Duration: 2 hours and 30 minutes

Weighting: 44%

Marks: 90

- Short-answer and extended-response questions on standard level and additional higher level material.

The questions on paper 2 test assessment objectives 1, 2 and 3.

The use of calculators is permitted. See the *Calculators guidance for examinations booklet* on the Programme Resource Centre.

Each student must have access to a clean copy of the *Physics data booklet* during the examination. It is the responsibility of the school to download a copy from IBIS or the Programme Resource Centre and to ensure that there are sufficient copies available for all students.

What about the IA?

There are four IA criteria for the scientific investigation. The marks and weightings are as follows.

Criterion	Maximum number of marks available	Weighting (%)
Research design	6	25
Data analysis	6	25
Conclusion	6	25
Evaluation	6	25
Total	24	100

The internal assessment requirements at SL and at HL are the same.

More about the IA soon enough...

Will I survive?

- ▶ Short answer: YES!
- ▶ The Diploma Programme is one of the most difficult high school programs in the world.
- ▶ DP Physics is seen as one of the most difficult courses offered by the DP.
- ▶ Physics is not required at our school, and the fact that you signed up shows that you are motivated to be here.
- ▶ Physics typically has one of the smallest class size (3-13 students at SH) -- this means more individual attention.
- ▶ We have some encouraging results from the past few years...

May 2025 Grade Boundaries

Subject: PHYSICS **Level:** HL

PAPER 1 (MCQ)			PAPER 1 B			PAPER TWO			PRACTICAL WORK			FINAL SUBJECT %		
Grade	From	To	Grade	From	To	Grade	From	To	Grade	From	To	Grade	From	To
1	0	9	1	0	2	1	0	7	1	0	3	1	0	13
2	10	11	2	3	5	2	8	15	2	4	6	2	14	22
3	12	14	3	6	7	3	16	24	3	7	10	3	23	33
4	15	18	4	8	9	4	25	36	4	11	13	4	34	45
5	19	22	5	10	12	5	37	48	5	14	16	5	46	58
6	23	26	6	13	14	6	49	60	6	17	19	6	59	70
7	27	39	7	15	20	7	61	90	7	20	24	7	71	100
Scaling factor = multiply by .615385			Scaling factor = multiply by .6			Scaling factor = multiply by .488889			Scaling factor = multiply by .833334					

May 2025 Grade Boundaries

Subject: PHYSICS **Level:** SL

PAPER 1 (MCQ)			PAPER 1 B			PAPER TWO			PRACTICAL WORK			FINAL SUBJECT %		
Grade	From	To	Grade	From	To	Grade	From	To	Grade	From	To	Grade	From	To
1	0	5	1	0	2	1	0	4	1	0	3	1	0	12
2	6	6	2	3	5	2	5	8	2	4	6	2	13	21
3	7	8	3	6	7	3	9	15	3	7	10	3	22	34
4	9	10	4	8	9	4	16	21	4	11	13	4	35	45
5	11	11	5	10	12	5	22	27	5	14	16	5	46	55
6	12	13	6	13	14	6	28	33	6	17	19	6	56	66
7	14	25	7	15	20	7	34	50	7	20	24	7	67	100
Scaling factor = multiply by .8			Scaling factor = multiply by .8			Scaling factor = multiply by .88			Scaling factor = multiply by .833334					

May 2025 Predicted vs Actual Grades (Paper 1A)

HL SL

Grade boundaries based on moderated mark

Grade	Minimum	Maximum
1	0	9
2	10	11
3	12	14
4	15	18
5	19	22
6	23	26
7	27	39
Moderated mark	Scaled mark	Grade
27	16.615	7
34	20.923	7
22	13.538	5

Grade boundaries based on moderated mark

Grade	Minimum	Maximum
1	0	5
2	6	6
3	7	8
4	9	10
5	11	11
6	12	13
7	14	25
Moderated mark	Scaled mark	Grade
10	8	4
4	3.2	1

May 2025 Predicted vs Actual Grades (Paper 1B)

HL SL

Grade boundaries based on moderated mark

Grade	Minimum	Maximum
1	0	2
2	3	5
3	6	7
4	8	9
5	10	12
6	13	14
7	15	20

Moderated mark	Scaled mark	Grade
17	10.2	7
17	10.2	7
13	7.8	6

Grade boundaries based on moderated mark

Grade	Minimum	Maximum
1	0	2
2	3	5
3	6	7
4	8	9
5	10	12
6	13	14
7	15	20

Moderated mark	Scaled mark	Grade
8	6.4	4
7	5.6	3

May 2025 Predicted vs Actual Grades (Paper 2)

HL SL

Grade boundaries based on moderated mark

Grade	Minimum	Maximum
1	0	7
2	8	15
3	16	24
4	25	36
5	37	48
6	49	60
7	61	90
Moderated mark	Scaled mark	Grade
63	30.8	7
72	35.2	7
39	19.067	5

Grade boundaries based on moderated mark

Grade	Minimum	Maximum
1	0	4
2	5	8
3	9	15
4	16	21
5	22	27
6	28	33
7	34	50
Moderated mark	Scaled mark	Grade
15	13.2	3
14	12.32	3

May 2025 Predicted vs Actual Grades (IA)

HL SL

Grade boundaries based on moderated mark

Grade	Minimum	Maximum
1	0	3
2	4	6
3	7	10
4	11	13
5	14	16
6	17	19
7	20	24
Moderated mark	Scaled mark	Grade
20	16.667	7
17	14.167	6
14	11.667	5

Grade boundaries based on moderated mark

Grade	Minimum	Maximum
1	0	3
2	4	6
3	7	10
4	11	13
5	14	16
6	17	19
7	20	24
Moderated mark	Scaled mark	Grade
12	10	4
9	7.5	3

May 2025 Predicted vs Actual Grades (Overall)

HL SL

**Predicted
grade**

Grade

7

7

6

7

5

5

**Predicted
grade**

Grade

4

4

2

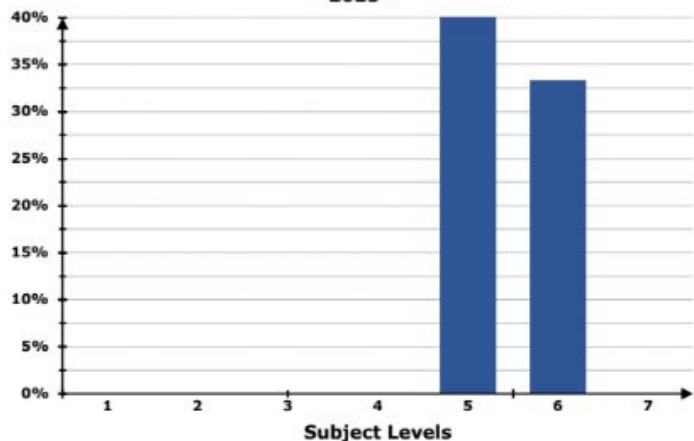
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May 2023 DP Results – Group 4 – Physics SL

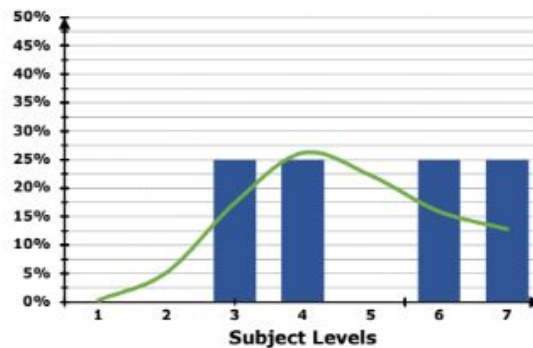
SUBJECT RESULTS FOR PHYSICS SL

Physics SL
- 2023 -

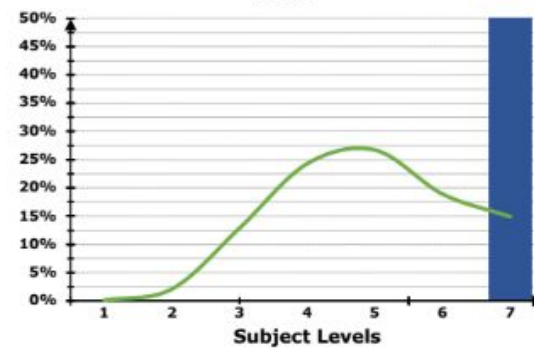


Students	3
Highest	6
Lowest	5
Median	5
School Average (SA)	5.33
World Average (WA)	
(SA - WA)	

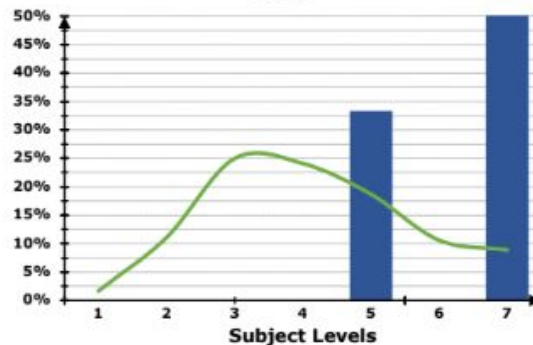
- 2022 -



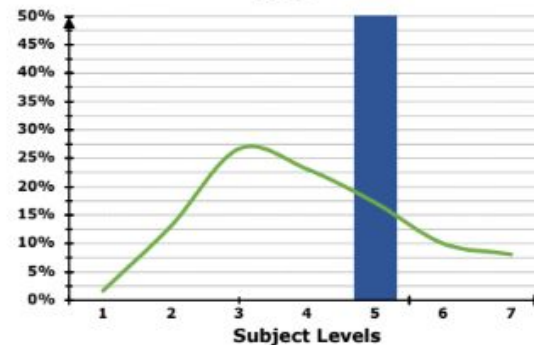
- 2021 -



- 2020 -



- 2019 -

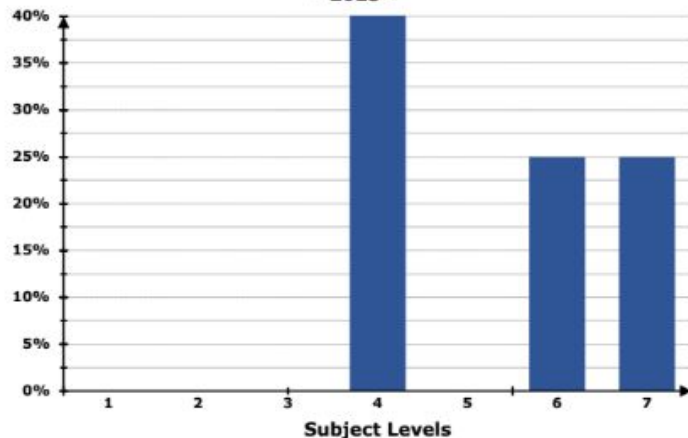




May 2023 DP Results – Group 4 – Physics HL

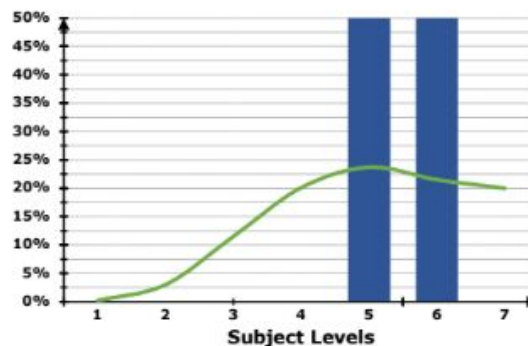
SUBJECT RESULTS FOR PHYSICS HL

Physics HL
- 2023 -



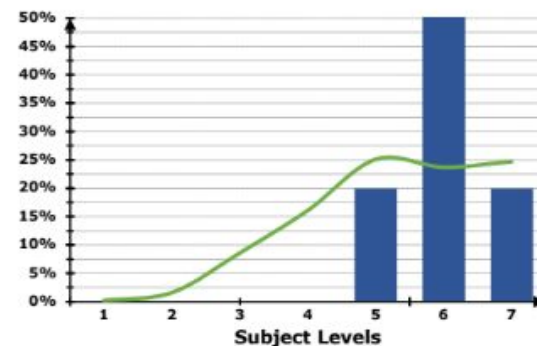
Subject Levels

- 2022 -



Subject Levels

- 2021 -



Subject Levels

no data for May 2020

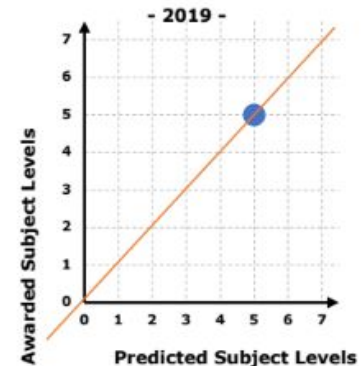
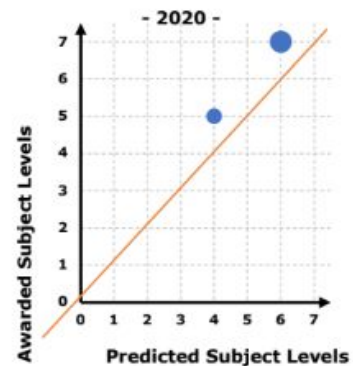
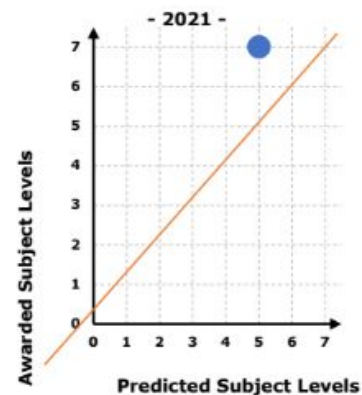
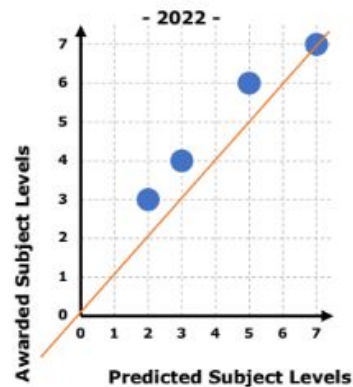
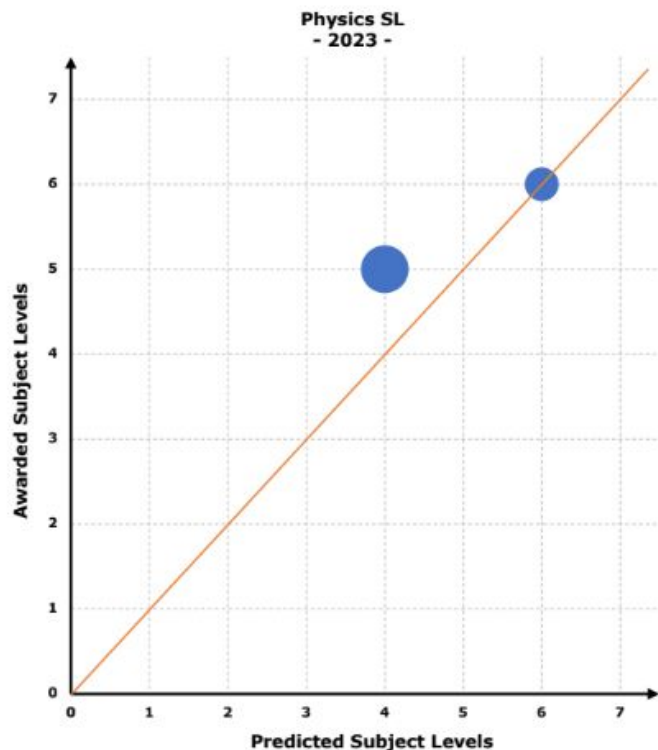
no data for May 2019

Students	4
Highest	7
Lowest	4
Median	6
School Average (SA)	5.25
World Average (WA)	
(SA - WA)	



May 2023 DP Results – Group 4 – Physics SL

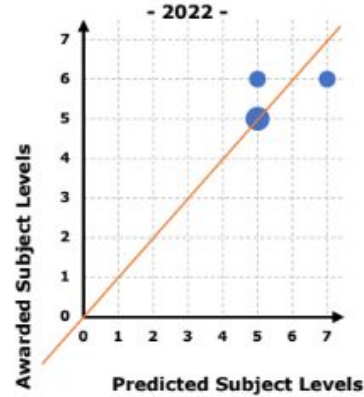
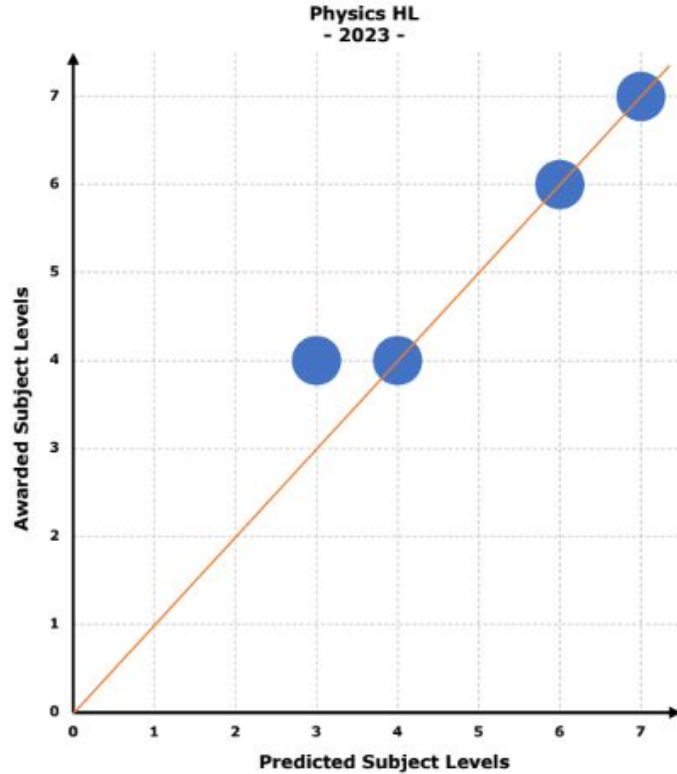
PREDICTED GRADES VS ACTUAL (AWARDED) GRADES FOR PHYSICS SL



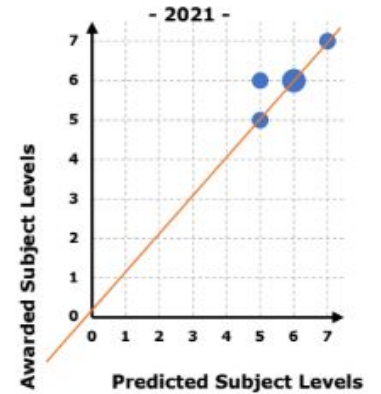


May 2023 DP Results – Group 4 – Physics HL

PREDICTED GRADES VS ACTUAL (AWARDED) GRADES FOR PHYSICS HL



no data for May 2020



no data for May 2019

Markbands 2024

Overall HL

Grade	Min	Max
3	24	37
4	38	49
5	50	60
6	61	72
7	73	100

Markbands 2023

Paper 1

HL

SL

Grade	Minimum	Maximum
1	0	10
2	11	11
3	12	14
4	15	18
5	19	21
6	22	25
7	26	40

Grade	Minimum	Maximum
1	0	7
2	8	9
3	10	11
4	12	13
5	14	16
6	17	19
7	20	30

Markbands 2023

Paper 2

HL

Grade	Minimum	Maximum
1	0	4
2	5	11
3	12	20
4	21	32
5	33	45
6	46	57
7	58	90

SL

Grade	Minimum	Maximum
1	0	2
2	3	6
3	7	12
4	13	19
5	20	25
6	26	32
7	33	50

Markbands 2023

Paper 3

HL

Grade	Minimum	Maximum
1	0	4
2	5	9
3	10	15
4	16	20
5	21	24
6	25	29
7	30	45

SL

Grade	Minimum	Maximum
1	0	3
2	4	7
3	8	10
4	11	14
5	15	19
6	20	23
7	24	35

Markbands 2023

IA

HL

SL

Grade	Minimum	Maximum
1	0	3
2	4	6
3	7	10
4	11	13
5	14	16
6	17	19
7	20	24

Grade	Minimum	Maximum
1	0	3
2	4	6
3	7	10
4	11	13
5	14	16
6	17	19
7	20	24

What to expect from Mr. Knight's class

▶ *Textbooks:*

- ▶ You will be given access to an online textbook through **Kognity.com**. Your progress will be tracked through your responses to in-text questions and assignments.
- ▶ You will be given a paper copy of the Oxford DP Physics study guide. Paper textbooks can be borrowed from the library.
- ▶ You will be given a 13-page **Data Booklet** identical to the one you will have access to during summatives -- **bring it to class everyday!**
- ▶ **You do not need to bring your physics textbook/workbook to class.**

▶ *Class Notes and Other Resources:*

- ▶ Class notes will be available to you in advance in PowerPoint form. As will other resources such as videos and online simulations. They will all be posted on our Organizing Document (OD) -- link on MB.

What to expect from Mr. Knight's class

► *Homework:*

- **Formative homework** will be assigned through Kognity throughout the course.
- **There is no formal due date for homework** – you are expected to complete it on your own and seek help if you need it.
- Mr. Knight is available in MYP Study Centre every Tuesday.
- You are strongly encouraged to complete old exam questions made by IB. Old exams and their markschemes (answer keys) are available to you on MB.
 - Note: The old exams correspond to the old curriculum, but there is a great amount of overlap between the old and new curricula.
- Partial and full lab reports will be due periodically throughout DP.

What to expect from Mr. Knight's class

- ▶ *Summatives:*
 - ▶ **Tests** take place during **and** after each of the topics we cover.
 - ▶ HL students will write an additional test that covers HL material.
 - ▶ Cumulative **mock exams** will take place during DP exam periods. **ALL PREVIOUSLY COVERED MATERIAL IS FAIR GAME.**
 - ▶ All tests/quizzes/mock exams will include:
 - ▶ Paper 1 type questions (multiple choice + data analysis)
 - ▶ Paper 2 type questions (short answer)
 - ▶ **All tests/quizzes/mock exams are comprised of questions from previous DP exams. Dozens of old exams are available to you on MB or elsewhere online.**
- ▶ *General Communication:* When in doubt, check the OD.

Intro to Physics



<https://www.youtube.com/watch?v=OoO5d5P0Jn4>

Topic 1 – Measurements and Uncertainty



<https://www.youtube.com/watch?v=urcQAKKAAI0>

Topic 2 – Mechanics



<https://www.youtube.com/watch?v=wGIZKETKKdw>

Topic 3 – Thermal Physics



<https://www.youtube.com/watch?v=vqDbMEdLiCs>

Topic 4 – Waves



<https://www.youtube.com/watch?v=Q3oltpVa9fs>

Topic 5 – Electricity and Magnetism



<https://www.youtube.com/watch?v=9k7zywli4Vg>

Topic 6 – Circular Motion and Gravitation



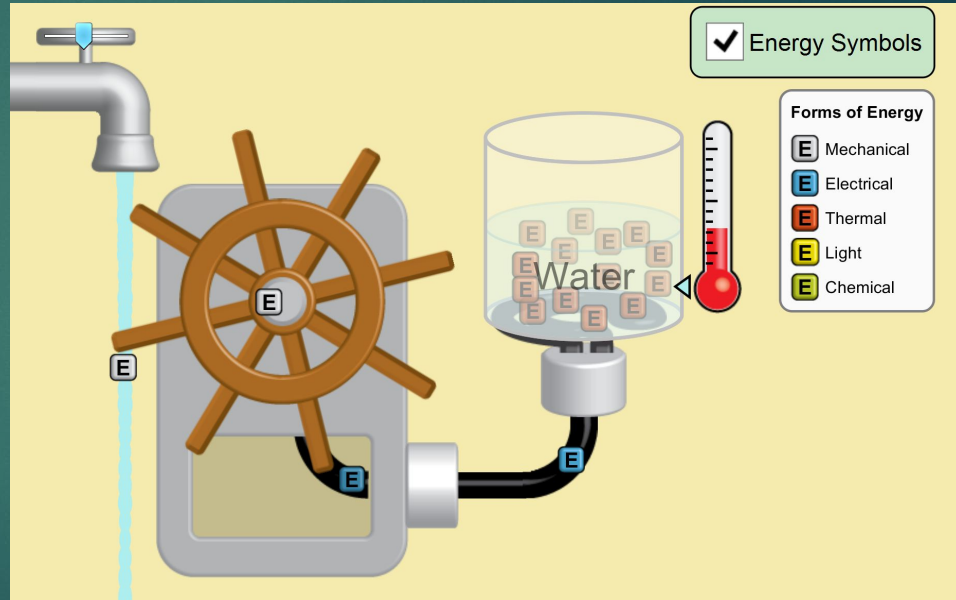
<https://www.youtube.com/watch?v=OTcdutlcEJ4>

Topic 7 – Atomic, Nuclear, and Particle Physics



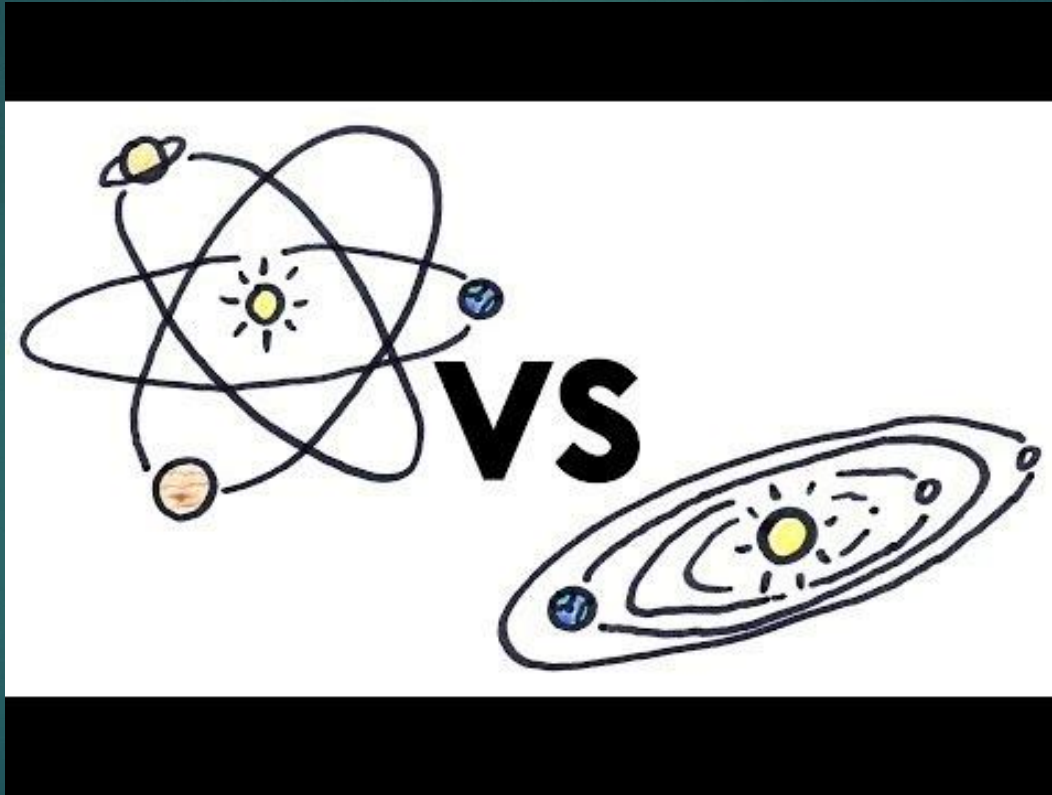
<https://www.youtube.com/watch?v=TRL7o2kPqw0>

Topic 8 – Energy Production



<https://phet.colorado.edu/en/simulation/legacy/energy-forms-and-changes>

Option D – Astrophysics



<https://www.youtube.com/watch?v=tmNXKgeUtJM>



**KEEP
CALM
AND
ENJOY
PHYSICS**